Qianyue Wang

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Education

South China University of Technology Sept. 2024–Jul. 2027
M.S. in Software Engineering, Advisor: Prof. Mingkui Tan

South China University of Technology Sept. 2020–Jul. 2024
B.Eng. in Software Engineering, GPA: 3.74/4.00
IELTS 7.0 (Listening 7.0 / Reading 8.0 / Writing 6.5 / Speaking 7.0)

Honors

National Scholarship	2025.10, 2023.10
Tencent Scholarship, Grand Prize	2025.10
Guangdong Province Society of Image and Graphics, Outstanding Service Award	2024.12
The 19th National Extracurricular Academic Science and Technology Contest, National Second Prize	2024.11
Outstanding Undergraduate Thesis	2024.06

Research Interests

My research focuses on **reliable and efficient LLM reasoning**. Starting from **model-centric test-time enhancement**, I found that static knowledge augmentation and additional computation alone cannot reliably handle errors and uncertainty emerging during reasoning, which motivated my shift toward **process-level interactive reasoning**, where external knowledge and human feedback dynamically guide and correct the reasoning process.

Selected Research Experience

Research Theme 1 Human–LLM Collaborative Reasoning and Interaction

- **Intervene When It Doubts: Conjunction-Guided Interactive Reasoning**[3]
  First Author · ICML 2026 May 2025–May 2026
  - **Problem**: Large reasoning models (LRMs) lack control over the depth and direction of reasoning, causing **overthinking** by redundant self-verification and **overshoot** by harmful post-solution revision.
  - **Contribution**: Revealed that transitional conjunctions mark **doubt-driven reasoning extensions** whose utility depends on the current reasoning state. Proposed **CGI**, a process-aligned human/LLM-in-the-loop framework that evaluates ongoing reasoning through rationality and completeness and injects actionable feedback to **dynamically control the reasoning depth and direction**. CGI improves AIME25 accuracy by **21.64%** while reducing reasoning length by **52.05%** over the base model.
- **Personalized Dual-track Intelligent Education System**
  Work in Progress Sept. 2025–Present
  - **Problem**: Current intelligent tutoring systems employ fixed teaching modes and explanation granularity, failing to adapt to students of **different and evolving proficiency levels**.
  - **Contribution**: Proposed a personalized dual-track intelligent education system equipped with a **step-level Socratic explanation mode** and a **discussion mode** for different learning stages. Combined with continuous tracking of student status, behavioral features, and knowledge mastery, the system **dynamically adjusts subsequent teaching strategies and explanation granularity**. Under medium-mastery student agents, the explanation-mode score rises from **5.09 to 7.61** and the discussion-mode score from **7.67 to 8.18**.

Research Theme 2 Controllable Content Generation

- **Generating Long-form Story Using Dynamic Outlining with Memory-Enhancement**[4]
  First Author · NAACL 2025 May 2024–Apr. 2025
  - **Problem**: Existing long-form generation methods struggle to balance **global plot planning** with **adaptive local creation**: fixed outlines cannot adapt to generation uncertainty, while incremental writing lacks plot completeness and accumulates long-range contextual conflicts.
  - **Contribution**: Proposed **DOME**, which couples **writing-theory-guided dynamic hierarchical planning** that expands fine-grained outlines during writing with **temporal-knowledge-graph memory** to aggregate long-range context. DOME **improves content diversity by 6.4%** and **reduces contextual conflicts by 53.7%** over a strong long-form generation baseline.
- **Industry–Academia Collaborative Research**: **Intelligent Main Grid Planning and Decision Support**
  Core Member ·Sponsored by China Southern Power Grid Technology Co., Ltd. Oct. 2025–Oct. 2028

- **Objective**: Develop an intelligent decision-support system that transforms **heterogeneous data from 500 kV main-grid planning schemes** into standardized analytical reports, supporting human experts in the **efficient, transparent, and controllable comparison and selection among candidate grid plans**. The collaboration is planned through **Oct. 2028**.
- **Contribution**: Developed **EvidFuse**, the multi-agent report-generation framework that decouples visualization construction from content organization and proposes the **write-time evidence construction** mechanism that enables**constructing and injecting real evidence into context during the writing process**. Established **traceable links** among source data, analytical conclusions, and generated content, and **LLM-assisted post-generation editing**. Figure-, section-, and report-level evaluations **outperform the baseline**.

Research Theme 3 Test-time Knowledge Enhancement

- **Trustworthy Government Q&A Technology Based on Large Language Model**[5]
  First Author · Journal of Software · Undergraduate Thesis Sept. 2023–Dec. 2024
  - **Problem**: Existing government Q&A systems struggle to generate responses that jointly satisfy **factual accuracy, regulatory compliance, and interpretability** in heterogeneous public-service scenarios.
  - **Contribution**: Built a multi-service government Q&A dataset, trained **GovGLM-6B**, and developed a trustworthy generation framework integrating **intent recognition** and **knowledge retrieval**, together with a **prototype system**. This work is the undergraduate thesis and was published in **Journal of Software**. GovGLM-6B outperforms Qwen-72B-Chat by **1.33% on BERTScore** and **6.74% on ROUGE-L**.
- **Preventing Overthinking in LRMs via Test-Time Precedent Learning**
  Work in Progress Dec. 2025 - Present
  - **Problem**: Large reasoning models repeatedly rediscover similar solution strategies at test time, causing redundant exploration, excessive token consumption, and unstable performance.
  - **Contribution**: Proposed **Precedent-Informed Reasoning (PIR)**, a test-time learning framework that retrieves relevant precedents and internalizes transferable reasoning strategies before solving the target problem, proactively reducing redundant exploration. On LiveCodeBench, PIR **improves accuracy by 11.61%** while **reducing reasoning length by 28.30%** over the base model.

Publications and Preprints

[1] **Qianyue Wang**, Jinwu Hu, Huanxiang Lin, Bolin Chen, Zhiquan Wen, Yaofo Chen, Yu Rong, Mingkui Tan. "Precedent-Informed Reasoning: Mitigating Overthinking in Large Reasoning Models via Test-Time Precedent Learning." *arXiv preprint arXiv:2602.14451*.

[2] **Qianyue Wang**, Huanxiang Lin, Jinwu Hu, Bailin Chen, Qing Du, Mingkui Tan. "EvidFuse: Writing-Time Evidence Learning for Consistent Text-Chart Data Reporting." *arXiv preprint arXiv:2601.05487*.

[3] **Qianyue Wang**, Jinwu Hu, Yufeng Wang, Zhiquan Wen, Yaofo Chen, Yu Rong, Mingkui Tan. "Intervene When It Doubts: Conjunction-Guided Interactive Reasoning." **ICML 2026**.

[4] **Qianyue Wang**, Jinwu Hu, Zhengping Li, Yufeng Wang, Daiyuan Li, Yu Hu, Mingkui Tan. "Generating Long-form Story Using Dynamic Hierarchical Outlining with Memory-Enhancement." **NAACL 2025**.

[5] **Qianyue Wang**, Jinwu Hu, Yufeng Wang, Yu Hu, Zhouqiang Qiu, Mingkui Tan. "Towards Trustworthy Government Q&A Based on Large Language Model." **Journal of Software**, 37(4):1740–1758, 2026.

[6] Yufeng Wang, Jinwu Hu, Ziteng Huang, Kunyang Lin, Zitian Zhang, Peihao Chen, **Qianyue Wang**, et al. "Enhancing User-Oriented Proactivity in Open-Domain Dialogues with Critic Guidance." **IJCAI 2025**.

[7] Shuhai Zhang, Zeng You, Yaofo Chen, Zhiquan Wen, **Qianyue Wang**, Zhijie Qiu, Yuanqing Li, Mingkui Tan. "Curse of High Dimensionality Issue in Transformer for Long-context Modeling." **ICML 2025**.

[8] Zhihui Zhang, Zhixuan Li, **Qianyue Wang**, Hanlin Wu, Manli Yang, Fengguang Zhao, Mingkui Tan, Shuangyan Han. "A Protein Fitness Predictive Framework Based on Feature Combination and Intelligent Searching." *Protein Science*, 33(12):e5211, 2024.

[9] Sijing Zhong, Jiayi Huang, Hengyu Meng, Zhuo Feng, **Qianyue Wang**, Zhengyu Huang, Lijie Zhang, Shiwei Gong, Weiyang Huang, Yusen Ying, Ning Li. "Machine Learning-assisted Performance Analysis of Organic Photovoltaics." *Materials Genome Engineering Advances*, 2(4), 2024.

Internship

Guangdong Artificial Intelligence and Digital Economy Laboratory Jun. 2025–Jun. 2026
Research Intern **Mentor**: Prof. Yuanqing Li

China Southern Power Grid Technology Co., Ltd. Jun. 2026–Sept. 2026
Research Intern **Mentor**: Eng. Liukai Chen